45

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION — SUMMER SEMESTER, 2023-2024

DURATION: 3 HOURS — FULL MARKS: 150

CSE 4849: Human Computer Interaction

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Suppose you are designing a mobile banking app and considering a specific task, "Transferring money to another account," to analyze five design principles of interaction design (visibility, feedback, consistency, constraints, and affordance).

   i. For each of the five design principles, write one sentence explaining how it applies to the money transfer task. 10 (CO3) (PO3)

   ii. Match each principle with the most relevant usability goal from the list: Effectiveness, Efficiency, Safety, Learnability, Memorability. Write one sentence justification for each goal. 5 (CO3) (PO3)

   iii. Explain in two sentences which user experience goals (one or more) from the list: satisfaction, enjoyment, trust, engagement, will be improved when these design principles are applied to the money transfer task. 5 (CO3) (PO3)

   b) In the context of conceptualizing interaction, an interactive system can be analyzed based on assumptions and claims. Write two assumptions about the user or system for the money transfer task, and two claims about what the system or design principles will ensure for the user while performing that task. 5 (CO3) (PO1)

2. a) What do you understand by divergent thinking and convergent thinking in the design process of a system using the double diamond model? Explain with a real-life example. 15 (CO3) (PO1)

   b) Social media app allows users to post updates, interact with friends, explore content, and receive notifications. For each task below, identify the interaction type (instructing, conversing, manipulating, exploring, responding) and justify your choice in one sentence. 10 (CO3) (PO1)

   - The user taps the Like button on a friends post.
   - The app sends a notification when someone comments on the users post.
   - The user scrolls through their feed to see new posts.
   - The user searches for a friend by typing their name in the search bar
   - The user sends a message to a friend using the chat feature.

3. a) Imagine you are designing a GPT-based educational assistant that helps university students prepare for exams. The system should collaborate with students by providing answers, explanations, and study guidance while ensuring students remain in control of their learning. Explain how the phases of user-centered design (UCD) can be applied, considering the inclusion of stakeholders in this system. 15 (CO3) (PO3)

   b) You are analyzing the design of an e-book reader system. The app supports several user tasks that are designed using different interface metaphors. For each of the following actions, identify the interface metaphor applied, and justify your choice in one sentence. 10 (CO3) (PO3)

   - A reader quickly jumps to the occurrence of a particular word inside the text without reading the whole book.

- The system displays a collection of available books in a grid, and the reader taps one to start reading.
- A user marks a passage of interest so they can return to it later or share it with others.
- When a reader logs into the app from a new device, all previously purchased and read books appear automatically.
- The reader expands a complex diagram on the page to view fine details more clearly.

4. a) To design a GPT-powered educational assistant, write the persona construction process based on the following steps. 15 (CO4) (PO1)
   - Identify behavioral variables
   - Map interview subjects to behavioral variables
   - Identify significant behavior patterns
   - Synthesize characteristics and relevant goals
   - Check for redundancy and completeness
   - Expand description of attributes and behaviors
   - Designate persona types

   b) Explain the relationship among persona, scenario, and goal with a real-life example. 10 (CO3) (PO1)

5. a) You are designing a Bangla mental health conversational AI system to provide empathetic support for the stakeholders. Answer the following:
      i. Identify three suitable data gathering methods, explain why each is appropriate, and indicate whether it collects qualitative or quantitative data. 9 (CO4) (PO1)
      ii. For each method, describe how you would record and analyze the data to inform system design, giving one example per method. 6 (CO4) (PO1)
      iii. Discuss two ethical considerations when collecting sensitive mental health data and how you would address them. 5 (CO4) (PO1)

   b) Differentiate between 'Evolutionary' prototyping and 'Throw-away' prototyping construction process. 5 (CO4) (PO1)

6. a) You are evaluating a Bangla mental health support chatbot. Participants must complete two fixed tasks with the chatbot:
   - Task 1: "Tell the chatbot you are feeling stressed."
   - Task 2: "Explain to the chatbot that you are feeling a bit lonely and ask it to suggest self-care activities."

   Two chatbot versions are tested: Version A (Empathetic) uses supportive and validating language; Version B (Neutral) provides only factual responses.
   Measures collected: completion time, error count, user satisfaction (1-5 scale), trust rating (1-5 scale).
      i. Identify the independent, dependent, control, confounding, and random variables in this empirical study. Give one line justification for each variable you have chosen. 10 (CO5) (PO3)
      ii. To compare two chatbot versions: Empathetic (supportive) vs Neutral (factual), decide whether a within-subjects experiment or a between-subjects experiment is more appropriate for this study. Justify your answer in terms of causal effect analysis and correlation analysis. 10 (CO5) (PO3)

   b) For the inspection-based evaluation of the mental health chatbot, which evaluation technique, between cognitive walkthrough and pluralistic walkthrough, would be appropriate? Justify your choice. 5 (CO5) (PO3)